

Dr. A. Chandrashekar

Associate Professor

Department of Mathematics, School of Mathematics and Computer Sciences
Central University of Tamil Nadu (CUTN), Thiruvarur – 610005, Tamil Nadu, INDIA

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Education

- **Ph.D. in Mathematics** July 2010
Indian Institute of Technology Madras (IIT Madras), Chennai, India
Thesis Title: *Standard and Semidefinite Linear Complementarity Problems*
- **M.Sc. in Mathematics** (81.87%) 2002–2004
Ayya Nadar Janaki Ammal College, Sivakasi (Affiliated to Madurai Kamaraj University)
- **B.Sc. in Mathematics** (82.45%) 1999–2002
Ayya Nadar Janaki Ammal College, Sivakasi (Affiliated to Madurai Kamaraj University)

Academic Positions & Research Visits

- **Associate Professor**, Department of Mathematics, Central University of Tamil Nadu Mar 2024 – Present
- **Assistant Professor**, Department of Mathematics, Central University of Tamil Nadu Mar 2012 – Mar 2024
- **Visiting Researcher**, Arbeitsgruppe Technomathematik, TU Kaiserslautern, Germany Jan 11 – Feb 25, 2023
- **Visiting Faculty**, Indian Institute of Management (IIM) Indore Nov 2013 – Jan 2014
- **Visiting Scientist**, Indian Statistical Institute (ISI), Hyderabad June 2013
- **Assistant Professor**, Dept. of Mathematics, BITS Pilani, K. K. Birla Goa Campus Dec 2011 – Feb 2012
- **Guest Faculty**, Department of Mathematics, University of Hyderabad Dec 2010 & Nov 2011
- **Research Associate (DST Postdoctoral Fellow)**, C R Rao AIMSCS, Hyderabad Feb 2010 – Nov 2011
- **Research Assistant (JRF)**, DST Project, Department of Mathematics, IIT Madras Sept 2004 – Dec 2004

Teaching Experience & Thesis Supervision

Teaching & Mentorship

- **Courses Taught:** Linear Algebra, Complex Analysis, Algebra I, Discrete Mathematics, Complex Analysis and Measure Theory, Functional Analysis, Optimization, Nonlinear Programming, Operations Research, Game Theory, Fixed Point Theory, Discrete Dynamical Systems, Scientific Computing Lab I (Python), and Scientific Computing Lab II (SageMath).
- **Teaching in MTTS:** Regular resident faculty since 2013 in the [Mathematics Training and Talent Search \(MTTS\)](#) programme (funded by NBHM, India). Student participant in 2002.
- **M.Sc. Project Guidance:** Supervision of final-year Integrated M.Sc. students at CUTN in independent reading, mathematical writing, and research dissertations.

Ph.D. Thesis Supervision

- **Dr. S. Gokulraj** Defense Date: Mar 16, 2021
Thesis Title: *A study on linear and bilinear games*
Current Position: Assistant Professor, National Institute of Technology Calicut, India.
- **Dr. A. Shanmugapriya** Defense Date: June 16, 2026
Thesis Title: *A study on the Lyapunov rank and the tensor product of cones*
Current Position: National Institute of Technology Puducherry, Karaikal, India.

- **Mr. Manu Mathew** 2021 – Present
Status: Ph.D. Candidate (Thesis ongoing).

Research Work & Publications

Preprints

1. Manu Mathew and **Chandrashekar A.** *On minimum enclosing angle of pointed polyhedral cones*, 2026.
2. **Chandrashekar A.** *Visualizing Riemann-Stieltjes integral as area under a curve*, 2025.

Refereed Journals & Book Chapters

1. Manu Mathew and **Chandrashekar A.** *Sufficient conditions for a doubly non-negative matrix to be completely positive*, to appear in *Computational and Applied Mathematics*, 2026.
2. Shanmugapriya A. and **Chandrashekar A.** *A construction of non-isomorphic polyhedral cones using Lyapunov rank*, *Linear and Multilinear Algebra*, 73 (2025), 2907–2916.
3. Shanmugapriya A. and **Chandrashekar A.** *Lyapunov-like transformations on the tensor product of nuclear pairs of proper cones*, *Electronic Journal of Linear Algebra*, 41 (2025), 142–151.
4. Manu Mathew and **Chandrashekar A.** *Self-duality and Maximal Angle of Proper Cones*, *Indian Journal of Pure and Applied Mathematics*, 56 (2025), 1177–1181.
5. Shanmugapriya A. and **Chandrashekar A.** *On the dual of the tensor product of semidefinite cones*, *The Journal of Analysis*, 32 (2024), 19–26.
6. Gokulraj S. and **Chandrashekar A.** *A characterization for the self-duality of proper cones*, *Positivity*, 27, 11 (2023).
7. Gokulraj S. and **Chandrashekar A.** *Linear Games and Complementarity Problems*, *Game Theory and Networks – New Perspectives and Directions* (2021), 17–35.
8. Gokulraj S. and **Chandrashekar A.** *Linear complementarity problems and bi-linear games*, *Applications of Mathematics*, 65 (2020), 665–675.
9. **Chandrashekar A.**, Sachindranath Jayaraman, and Vatsalkumar N. Mer. *A characterization of non-negativity relative to proper cones*, *Indian Journal of Pure and Applied Mathematics*, 51(3) (2019), 935–944.
10. Gokulraj S. and **Chandrashekar A.** *On symmetric linear games*, *Linear Algebra and its Applications*, 562 (2019), 44–54.
11. **Chandrashekar A.**, Sachindranath Jayaraman, and Vatsalkumar N. Mer. *Semipositivity of linear maps relative to proper cones in finite dimensional real Hilbert spaces*, *Electronic Journal of Linear Algebra*, 34 (2018), 304–319.
12. **Chandrashekar A.** and Sachindranath Jayaraman. *Linear maps on $M_n(\mathbb{R})$ preserving Schur stable matrices*, *Bulletin of Kerala Mathematical Association*, 16 (2018), 127–138.
13. **Chandrashekar A.** and Sachindranath Jayaraman. *Full column rank preservers that preserve semipositivity of matrices*, *Special Matrices*, 6 (2018), 37–45.
14. **Chandrashekar A.** and Sachindranath Jayaraman. *Diagonalizability of the product versus diagonalizability of the individual factors*, *Functional Analysis, Approximation and Computation*, 8(3) (2016), 31–35.
15. **Chandrashekar A.**, Parthasarathy T., and Vetrivel V. *Solving strongly monotone linear complementarity problems*, *International Game Theory Review*, 15(4) (2013), 1340035.
16. **Chandrashekar A.**, Parthasarathy T., and Ravindran G. *On strong Z matrices*, *Linear Algebra and its Applications*, 432 (2010), 964–969.
17. **Chandrashekar A.**, Parthasarathy T., and Vetrivel V. *On the P'_2 and P_2 -properties in the semidefinite linear complementarity problem*, *Linear Algebra and its Applications*, 432 (2010), 134–143.

Sponsored Research Projects

- **Early Career Research Award (ECRA)** – *Linear complementarity problems over the cone of completely positive matrices*
Funding Agency: Science and Engineering Research Board (SERB), India 2018–2021
Sanctioned Amount: Rs. 21,22,560/- (Completed)
- **Mathematical Research Impact Centric Support (MATRICS)** – *A study on proper cones with applications to mathematical programming*
Funding Agency: Science and Engineering Research Board (SERB), India 2020–2023
Sanctioned Amount: Rs. 6,60,000/- (Completed)

Professional Activities & Events Organized

Recent Invited Talks & Lectures

- **Invited Symposium Talk:** *Completely Positive Matrices*, Symposium on Matrix Analysis, 41st Annual Conference of Ramanujan Mathematical Society (RMS 2026), Manipal Institute of Technology Bengaluru July 2–5, 2026
- **Resource Person:** MTTS 2026 lectures at Indian Institute of Space Science and Technology (IIST), Thiruvannamalai May–June 2026
- **Resource Person:** *Initiation into Mathematics* programme, Sri G.V.G. Visalakshi College for Women, Udumalpet Mar 17–23, 2025

Conferences & Workshops Organized at CUTN

- [Mathematics Training and Talent Search \(MTTS\)](#), funded by NBHM May 20 – June 15, 2024
- [Pedagogical Training for Mathematics Teachers \(PTMT\)](#), funded by NBHM Dec 11–16, 2023
- Workshop on Differential Geometry (Speaker: Prof. S. Kumaresan), funded by CUTN Mar 11–16, 2022
- Workshop on Game Theory (Speaker: Prof. T. Parthasarathy), funded by SERB Feb 17–18, 2022
- [Pedagogical Training for Mathematics Teachers \(PTMT\)](#), funded by NBHM Dec 17–29, 2018
- Advanced Instructional School on Mathematical Programming, funded by NCM July 4–23, 2016
- Mathematics Workshop for Undergraduate Students, funded by NBHM Jan 23 – 25, 2016
- Workshop on Analysis and Geometry, funded by CUTN Feb 13–14, 2015
- Mathematics Workshop for Undergraduate Students, funded by NBHM Jan 30 – Feb 1, 2015

Administrative & Governance Roles

- **University Level:** Member, The Court (August 2025); Coordinator, Remedial Coaching Cell (Nov 2015–Nov 2025); Member, Academic Council, Library Advisory Committee, Hostel Committee, and Staff Grievance Redressal Committee.
- **Department Level:** Head i/c (Dept. of Mathematics), Board of Studies (BoS), Admission Committee, and Coordinator, Astronomy Club.
- **External Institutional Roles:** Member, Academic Council, Sri G.V.G. Visalakshi College for Women, Udumalpet Academic Years 2024–2027

MOOCs & E-Content Development

- 73-lecture series (7 modules) for *A Foundation Course in Mathematics*, broadcast on SwayamPrabha DTH Channel (Language: Tamil).
- Full DTH course on *Operations Research* for SwayamPrabha DTH Channel Sept 2025

Memberships, Honors & Awards

Professional Memberships

- Special Invitee, Mathematics Training and Talent Search (MTTS) Trust Apr 2021 – Present
- Member, Institute for Operations Research and the Management Sciences (INFORMS), USA 2022–2023
- Member, American Mathematical Society (AMS), USA 2021–2022
- Life Member, Ramanujan Mathematical Society (RMS), India
- Life Member, IIT Madras Alumni Association

Honors & Fellowships

- **Senior Research Fellowship (SRF)**, CSIR, India (at IIT Madras) Feb 2007 – Jan 2010
- **Junior Research Fellowship (JRF)**, CSIR, India (at IIT Madras) Jan 2005 – Jan 2007
- **NBHM Research Award (JRF)**, National Board for Higher Mathematics 2004
- **NBHM M.Sc. Fellowship**, National Board for Higher Mathematics 2002
- **INSA Summer Research Fellow**, University of Hyderabad (Guide: Prof. V. Kannan) Apr–May 2002

Community Service & Academic IT Initiatives

- **Peer Reviewer:** Active referee for several peer-reviewed international journals and AMS Mathematical Reviews.
- **MOODLE LMS Administration:** Setup and management of the [CUTN Learning Management System](#) (2013–present) and [MTTS Classrooms](#).
- **KOHA LMS Administration:** Setup and management of the [CUTN Library Management System](#) during its initial setup (2012–2017).
- **Open Source Software Promotion:** Invited talks on Python, SageMath, and GeoGebra across colleges in Tamil Nadu; introduced M.Sc. lab courses in Python and SageMath at CUTN.